

Train-Test overlap:

- + ML 101: don't train on test data
- + pre-foundation models (ImageNet, SQuAD): well-defined train-test split.
- + Norway: train on Internet, but don't tell people your data

so + solution 1: try to infer train-test overlap ^{from} ~~from~~ model

solution 2: encourage reporting norms (e.g., people report confidence intervals)
model providers should report train-test overlap

Summary:

pre-foundation models: we evaluate methods (standardized train-test
encourage algorithmic ^{splits} innovation from research

foundation models: we evaluate models / systems anything goes
evaluating models/systems is useful for downstream users

Either way, we need to define rules of the game!

takeaways:

- #1: There is no one perfect evaluation, choose the eval depending on what you're trying to measure.
- #2: ~~Always~~ Always look at the individual instances and the predictions.
- #3: There are many aspect to consider: capabilities, safety,
- #4: clearly state rules of games: (methods vs ^{models} systems) costs, realism.

Lecture #13: Training Data Curation

(# previous lectures: how to train a model given data

Next two lectures: what data should we train on?

data is the most important for training

introduction: companies for open-sources models only open-weight, open-architecture, open-training procedure,

But: no less info on Data

why?: ~~get~~ competitive dynamics
copyright liability / responsibly use data: Don't get sued

⇒ supervised learning requires heavy data annotation, but now few annotation needed, but still needs data curation/cleaning

stages of training: {
#1: pre-training on raw text.
#2: mid-training on some high-quality data to enhance capabilities
#3: post-training: fine-tune on instruction following data or reinforcement learning)

#1 + #2 ⇒ Base model

#3 ⇒ instruct / chat model

Do we need large amount of low-quality data or small amount of high-quality data?